

Sequences and series of functions

Math 163 • Prof. Chowdhury

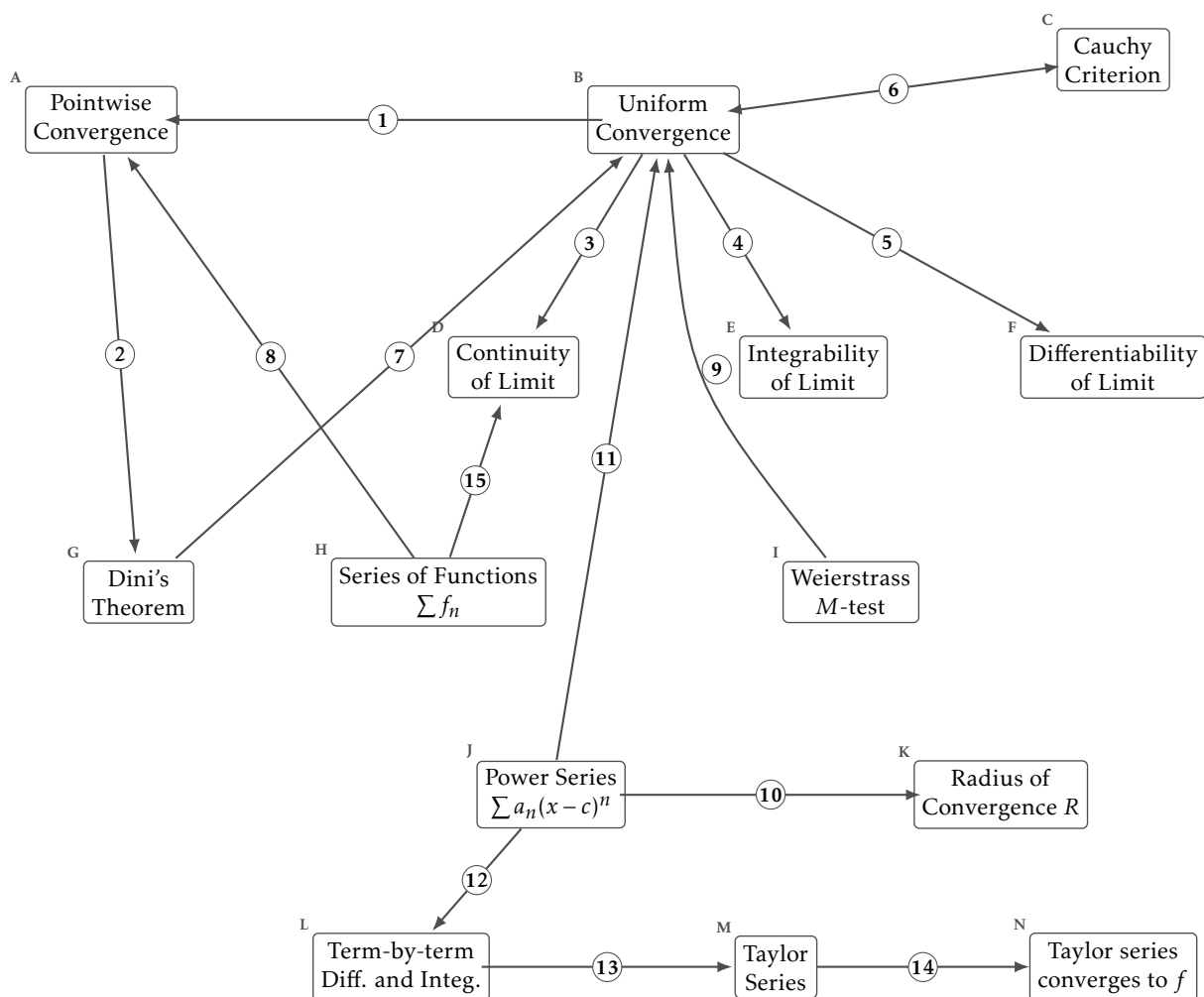
The next page has 14 ideas as boxes, with 15 numbered arrows between them and nothing written on any arrow. Every arrow is a relationship that holds in the direction drawn, and it does not have to be an implication. Sometimes one property forces another. Sometimes one box is defined in terms of the other. Often one box is the tool that settles the other: the integral test doesn't imply anything about the p -series, it is what pins down where the p -series turns.

For each numbered arrow, say in one sentence what the relationship is. Name the theorem where there is one, state the hypothesis doing the work, or say which box settles which. An arrow with a head at each end runs both ways.

When you have had a go at all of them, check yourself at [subhadipchowdhury.github.io/teaching/labs/maps/functions-sequences/](https://github.com/subhadipchowdhury/teaching/labs/maps/functions-sequences/). That page draws the same map and will show you the answers one arrow at a time, or all at once.

What each box means

- A. Pointwise Convergence.** For each x and each $\varepsilon > 0$ there is an N depending on both, with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$.
- B. Uniform Convergence.** For each $\varepsilon > 0$ there is an N depending only on ε , with $|f_n(x) - f(x)| < \varepsilon$ for $n \geq N$ and every x at once.
- C. Cauchy Criterion.** $\{f_n\}$ converges uniformly exactly when for each $\varepsilon > 0$ there is an N with $|f_m(x) - f_n(x)| < \varepsilon$ for all $m, n \geq N$ and all x .
- D. Continuity of Limit.** Each f_n continuous and $f_n \rightarrow f$ uniformly gives f continuous.
- E. Integrability of Limit.** Each f_n integrable and $f_n \rightarrow f$ uniformly gives f integrable, with $\lim \int f_n = \int \lim f_n$.
- F. Differentiability of Limit.** $f_n \rightarrow f$ pointwise together with $f'_n \rightarrow g$ uniformly gives $f' = g$. It is the derivatives that have to converge uniformly.
- G. Dini's Theorem.** $f_n \rightarrow f$ pointwise on $[a, b]$, monotonely, with every f_n and f continuous, gives $f_n \rightarrow f$ uniformly.
- H. Series of Functions** $\sum f_n$. $\sum f_n$ converges when its partial sums $S_N(x) = \sum_{k \leq N} f_k(x)$ do. A series of functions is a sequence of functions.
- I. Weierstrass M-test.** If $|f_n(x)| \leq M_n$ for every x and $\sum M_n < \infty$, then $\sum f_n$ converges uniformly and absolutely.
- J. Power Series** $\sum a_n(x - c)^n$. A series of functions whose n th term is a monomial of degree n centred at c .
- K. Radius of Convergence** R . $\sum a_n(x - c)^n$ converges absolutely for $|x - c| < R$ and diverges for $|x - c| > R$. The endpoints are checked one at a time.
- L. Term-by-term Diff. and Integ..** Inside $(c - R, c + R)$ a power series can be differentiated and integrated term by term, and the result has the same radius R .
- M. Taylor Series.** $\sum f^{(n)}(a)(x - a)^n/n!$. The coefficients are forced by differentiating the power series at its centre.
- N. Taylor series converges to f .** The Taylor series converges to $f(x)$ exactly when the remainder $R_N(x) = f(x) - P_N(x)$ tends to zero.



The arrows

- | | |
|---------|----------|
| 1. | 9. |
| 2. | 10. |
| 3. | 11. |
| 4. | 12. |
| 5. | 13. |
| 6. | 14. |
| 7. | 15. |
| 8. | |

Counterexamples and benchmarks

- x^n on $[0, 1]$ takes continuous functions to a discontinuous limit, pointwise.
- $\sqrt{x^2 + 1/n^2} \rightarrow |x|$ uniformly, and the limit is not differentiable at 0.
- $(1/n)\sin(n^2x) \rightarrow 0$ uniformly while its derivatives diverge.
- $x/(1 + n^2x^2) \rightarrow 0$ uniformly, and the derivative of the limit is not the limit of the derivatives at 0.
- e^{-1/x^2} extended by 0 is infinitely differentiable at 0 with every Taylor coefficient zero, so its Taylor series converges and not to it.

One last question

Which single arrow carries the most of this chapter? One quantifier separates the two kinds of convergence, and most of the chapter follows from where it lands, so say whether your answer is the arrow that moves it.

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